

Harsh Tiwari

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EXPERIENCE

WellnessLiving

Senior Software Engineer

Bengaluru, India

April 2025 - Present

- Engineered a **multi-tenant Voice AI backend** with a generalized **function-calling layer** to orchestrate **LLM-driven actions** across provider integrations, enabling extensible tenant-specific workflows through **RAG** and tool-execution pipelines.
- Re-architected **Cerebrum**, our LLM microservice, to deliver **sub-second telephony responses** by streaming model output, parallelizing retrieval and tool preparation, trimming prompt-construction overhead, and removing synchronous bottlenecks from the inference path.
- Scaled a **WebSocket platform** with stateless services, **JWT authentication**, and rate limiting, improving connection stability and reducing customer-reported issues in real-time chat.
- Operationalized **Airflow-based ingestion pipelines** from systems including **Intercom** and **Zoom** into **Redshift**, improving the reliability of centralized analytics and downstream reporting.
- Launched a **RAG-backed analytics interface** with **SQL agents** to enable **self-serve reporting** for internal teams and reduce dependence on manual data requests.
- Implemented **stage-level latency instrumentation** across **STT**, request assembly, **LLM inference**, tool execution, and **TTS** to attribute tail latency in the telephony path and drive **sub-second response optimizations** in production voice systems.

Kusho

Full Stack Developer

Bengaluru, India

August 2023 - April 2025

- Built **Redis- and RQ-based worker pipelines** for scalable **LLM task execution**, prompt processing, and **asynchronous job orchestration** across large API payloads, improving throughput under heavy background workloads.
- Designed a unified **service-definition DSL** for **Postman**, **OpenAPI/Swagger**, **GraphQL**, and **Protocol Buffers**, then integrated **LLM APIs** to auto-generate test cases and reduce test authoring time.
- Built an **API testing and execution platform** with **50+ workflow node types** for end-to-end orchestration, improving large-flow execution performance and shortening test runtime for complex workflows.
- Developed a **Chrome extension** that records user interactions and generates **Playwright test scripts**, implemented robust selectors with fallback strategies for UI changes, and delivered **95% stability**, **3x test coverage**, and **70% less manual QA**.
- Built a **distributed test execution engine** that modeled large API collections as **DAGs**, executing nodes via **Redis-backed worker queues (RQ)** with dependency-aware scheduling, parallel fan-out, and isolated execution pipelines to support high-concurrency workflow simulation across **1000+ businesses**.
- Built a **compiler** that normalized **Postman**, **OpenAPI**, **GraphQL**, and **Protobuf** specifications into a common executable graph IR, enabling a **shared runtime** for test generation, validation, and large-scale API workflow execution.

Castler

Full Stack Developer

Delhi, India

May 2021 - August 2023

- Built an **event-driven payment platform** for **UPI/IMPS** using message queues, idempotency controls, and circuit breakers, processing **INR 50Cr+ monthly** across **2M+ transactions** with **sub-200ms latency** while reducing infrastructure cost by **35%**.
- Architected a **fault-tolerant payee validation pipeline** that absorbed **1000+ RPM peak demand** against a heavily rate-limited upstream bank API, using asynchronous queueing, deduplication, backpressure, and stateful retries to avoid user-facing failures during rush-hour traffic.
- Led a **zero-downtime migration** of **200GB** of financial data from **MongoDB** to **MySQL** using blue-green deployment and dual-write architecture, then built **Airflow reconciliation pipelines** that improved data consistency, delivered **40% faster queries**, and reduced infrastructure cost by **25%**.
- Implemented **OAuth 2.0/PKCE authentication** and automated security scanning with **SonarQube**, **Snyk**, and **OWASP ZAP** to strengthen application security, release hygiene, and compliance readiness.
- Built **automated state-reconciliation and recovery workflows** across gateway responses, asynchronous bank callbacks, and internal ledger state to detect mismatched transaction states, recover failed payment flows, and preserve **financial correctness at scale**.

SKILLS

Languages: Python, TypeScript, Go, SQL

Backend & APIs: Node.js, NestJS, REST APIs, GraphQL, Microservices, Event-Driven Architecture, Distributed Systems

Data & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Airflow, AWS, Docker, Kubernetes, CI/CD

AI/ML: LLM APIs, RAG, Vector Databases

EDUCATION

Amity University

Bachelor of Technology in Computer Science

Lucknow, India

Aug. 2017 – May 2021